



SECTION A

General Unit Information:

Unit Name	Desktop Applications Development And Software Engineering
Unit Code	CIS116-2
Level	5
Credit Value	30
Location(s) of Delivery	On Campus and Off Campus
Period(s) of Delivery	15 weeks
Pre-requisites or Restrictions	N/A
Aims and Relevance	This unit aims to help you: • To develop desktop applications in an object-oriented programming language with object-oriented analysis and design; • To understand the process of modelling real world problems and systems in a programming language; • To develop the students' knowledge and understanding of all aspects of the software development process from an engineering viewpoint; • To develop the students' ability to analyse and apply a diverse range of concepts, principles and techniques to the whole software life cycle; • To extend your knowledge and skills in advanced programming and software development methods
Syllabus Content	This unit emphasizes the need for you to acquire in-depth programming skills for developing desktop applications, within the context of the Object-Oriented paradigm. Furthermore, you are encouraged to contextualize the programming skills within the software engineering life cycle (so that you will see the "forest" not only the "trees"). The entire life cycle is covered (from requirements gathering, through analysis, design, implementation, and testing). The emphasis throughout is upon critical analysis and reflection, and problem solving within realistic settings and scenarios of relevance to industry and commerce.

Learning Outcomes

On completion of this unit you should be able to

1. Demonstrate the following knowledge and understanding

Systematically categorize the concepts, principles and techniques for object-oriented desktop applications development and software engineering processes.

2. Demonstrate the following skills and abilities

Utilise object-oriented analysis and design for developing desktop applications while applying a range of concepts, principles and techniques to the software development processes and making use of well-chosen object-oriented analysis and design techniques.

PSRB Outcomes and/or Apprenticeship Standard

N/A

Summary Learning Hours

	Scheduled	Guided	Independent	Autonomous	Placement	Total
Hours	72	75	153	0	0	300
Percentage	24%	25%	51%	0%	0%	100%

Approach to Learning

The unit consists of Lectures, Practical Sessions, Seminars and Independent Learning

Lectures:

Used to transfer and contextualise theoretical concepts relating to object oriented programming and software engineering. Students may also test out ideas in practice within the context of a lecture. Not just about sitting and listening but often within an interactive in approach.

Practical Sessions:

Designed to develop skills through the application of object oriented programming concepts in practice and to develop a practical understanding of the field through hands-on application development.

Seminars:

Designed to involve you in discussion and detailed consideration of software engineering concepts, ideas and performance practice. Some seminars may also involve in-class discussion to both exemplify and develop understanding on a particular topic.

Independent Learning:

This is a crucial part of developing as a learner. Independent Learning is designed to facilitate student's independence through enabling them to take responsibility for your learning. Students will be required to prepare work outside the class/lecture contact time and should regard this in the same way as preparation and research towards any assignment. The practice of developing an independent, self-motivated approach to learning and delivery of your work is essential to student's success and to achieving good marks.

Assessment Summary

No	Assessment Method	Learning Outcome(s)	Weight	Submission Week	Length of Assessment
1	Coursework - Artefact	1 and 2	30	7	na
2	Ex	1 and 2	70	13	2 hours

Learner Development

This Unit particularly focuses on the development of your abilities in the following areas:

Enquiry	You are actively engaged in enquiry during the learning process and you have to articulate your own understanding of the issues examined in the class and/or in the exam.
Contextual Understanding, including where employer based	This unit allows you to consider the impact of object oriented programming and software engineering across a number of domains. It also fosters understanding of future applications and technologies.
Collaboration	You will be engaged in discussion within informal peer groups in the seminars for knowledge you learnt in this unit.
Enterprise	The learning outcomes in this unit come from working on given problems related to the real-world and/or enterprise context. They will challenge the students to make judgments not in their previous experience.
Assessment 1 Details	You will submit an artefact (piece of object-oriented software) as specified in the format outlined in the detailed assessment brief for the unit. The artefact produced will relate to development of software for a desktop environment
Assessment 2 Details	You will undertake a closed-book examination of 2 hours duration. The examination will cover key topics in the area of software development.
Assessment 3 Details	
Assessment 4 Details	

Threshold Expectations

In order to pass Assessment 1 you will need to :

Apply the object-oriented paradigm to design a solution for a given scenario

Utilise your knowledge of C# within the .NET framework to conceptualise a solution for a given scenario.

Express an understanding of software testing through the utilisation of a test-harness to assess the logical correctness of a desktop application.

In order to pass Assessment 2 you will need to:

Explain fundamental concepts used in developing desktop applications with the object-oriented paradigm.

Apply sorting and searching algorithms used in software development within a given programming language.

Identify key strategies used within Software Development and apply them correctly to problem solving questions.

In order to pass Assessment 3 you will need to:

In order to pass Assessment 4 you will need to:

Section B

Recommended Reading

Core Text - this unit is supported by the following core text:

See <https://rl.talis.com/3/bedfordshire/lists/4D608C77-6955-19FB-F607-291183D2821B.html?lang=en-GB>

Guided Reading

The following is expected reading for this unit. Details of what to read and when will be provide in the BREO site.

See <https://rl.talis.com/3/bedfordshire/lists/4D608C77-6955-19FB-F607-291183D2821B.html?lang=en-GB>

Independent Study

To receive high grades you will need to demonstrate your wider reading. The following resources provide useful background reading for the material in this unit. This is not an exhaustive list and students should read widely from the variety of journals available in the Learning Resources Centre.

See <https://rl.talis.com/3/bedfordshire/lists/4D608C77-6955-19FB-F607-291183D2821B.html?lang=en-GB>

Equality Impact Assessment

Question	Y/N/NA	Additional anticipatory adjustments/actions if necessary
Learning materials will be made available in advance of sessions for students to adapt as appropriate?	Y	Materials will be made available for students online in advance before the scheduled delivery
The approach to teaching and learning is sufficiently flexible to enable all students to succeed?	Y	
The approach to group work takes account of the needs of students with disabilities and from diverse backgrounds?	NA	No group work assessment is required in this unit.
The approach to practical work takes account of the needs of students with disabilities?	Y	
Students with a protected characteristic* have an equal opportunity to achieve the learning outcomes?	Y	
The assessment tasks provide all students with an equal opportunity to succeed?	Y	

Any other aspects of the unit that might pose potential challenges from an equality or diversity perspective have been considered?	NA	
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* Age, Gender reassignment, Marriage and civil partnership, Pregnancy and maternity, Race, Religion and belief, Sex, Sexual orientation

Section C

Administrative Information – Faculty completion	
Faculty	Creative Arts Technologies and Science
Portfolio	CU-COM
School/Department	School of Computer Science and Technology
HECOS Code(s)	100374
Unit Co-ordinator	Renxi Qiu
Apprenticeship ?	No

Courses that this unit is delivered on
BSc (Hons) Artificial Intelligence and Robotics
BSc (Hons) Software Engineering
BSc (Hons) Computer Science
BEng (Hons) Computer Systems Engineering